

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, Go, Java, JavaScript, TypeScript, C++, SQL, Bash

Backend & APIs

REST APIs, microservices, Node.js, Express, FastAPI, Fastify, multi-tenant services, async services, integration testing

Distributed Systems

Spark / PySpark, distributed data processing, multithreading, load balancing, Hadoop / MapReduce, Linux profiling

Cloud & DevOps

Linux, Docker, Kubernetes, CI/CD, GitHub Actions, AWS, GCP, Azure

Databases & Cache

MongoDB, DynamoDB, Redis, PostgreSQL, SQLite

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Engineered a **scalable multi-tenant NHS eTriage backend** for **8+ UK clients** (Node.js, MongoDB) processing **10,000+ monthly submissions** as a production-grade system integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines, contributing to **\$340K+** in revenue.
- Architected a full-stack data annotation platform (**React, Node.js, MongoDB**) used daily by the AI team to label data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual operations by **100+ hours/month** and improved labeling throughput by **60%**.
- Acted as **primary technical point of contact** for the eTriage backend, owning it end-to-end across feature releases and client integrations: monitored production logs to triage tenant-specific incidents, hardened authentication and rate limiting, and ran knowledge transfer to onboard a second developer to on-call readiness in **~6 weeks**.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated the core codebase from **JavaScript to TypeScript** across multiple repositories, improving type safety, software design, and long-term maintainability across services.

Projects

Spark ETL Optimization - Distributed Processing at ~1.4B Rows

github.com/Deva-1903/Spark-ETL-Optimization

Python, Apache Spark / PySpark, Spark UI, SQL

- Cut end-to-end Spark ETL runtime **~25%** (17 to 19 min down to 14.3 min) on a **~1.4B-row** NYC TLC dataset by replacing shuffle joins with broadcast joins (minutes to ~0.05s), enabling Adaptive Query Execution and skew-join handling, and tuning shuffle partitions, schema-aware reads, and year/month partitioning.
- Reduced analytics tail-latency ratio (P99/P50) from **2.14x to 1.74x** via partition and column pruning, filter pushdown, and pre-computed derived columns; used Spark UI stage-level timings to localize shuffle and skew bottlenecks and document a repeatable optimization playbook.

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a **Python/FastAPI service** for a multi-stage agentic web-discovery pipeline that plans typed retrieval, reranks pages, classifies evidence regimes, extracts entities with deterministic parsers before LLM fallback, verifies outputs, and returns cell-level provenance for every result.
- Hardened the service with **150+ automated tests**, provider routing and fallback across LLM backends, and pipeline instrumentation to keep extraction behavior predictable across backend and tool failures.

cf-ai-research-scout - Edge-Native AI Research Assistant

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers, WebSockets, Durable Objects, SQL

- Built an edge-native AI research assistant on Cloudflare Workers: agent fetches web sources, generates multi-step research digests, and **streams responses to the client over WebSocket** in real time, prototyped end-to-end with embedded SQL storage.
- Designed **durable workflow boundaries** so multi-step tasks survive process restarts and individual tool failures; agent state and conversation history persist server-side without an external database.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.